

A168 · HyperFlow: Hypergraph-Based Flow-Matching for Protein Conformation Generation

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Session: Day 1 · Thu Oct 1, 2026 · 2:15–4:15 PM

Protein function emerges from its sequence and structural dynamics, where collective motions of multiple residues constitute and connect distinct structural states. These conformational ensembles and their functional properties cannot be fully captured by a single static structure. Current generative models attempt to predict such distributions from one structure and reproduce per-residue flexibility well. However, conditioning on a single structure while reconstructing individual conformers favors the conditional mean, causing distinct states to collapse into a smoothed mode. We introduce HyperFlow, a flow-matching generative model that combines higher-order geometric representations with distribution-level supervision. HyperFlow represents proteins as hypergraphs, enabling the model to capture collective multibody interactions that drive conformational transitions. A distribution-level objective then encourages these motions to generate distinct modes of the conformational ensemble rather than average structures. On the ATLAS dataset, HyperFlow matches state-of-the-art performance in per-residue flexibility (RMSF $r = 0.88$) while increasing collective-mode similarity to 56%, compared with 40–48% for AlphaFlow, ESMFlow-T, and BBFlow. HyperFlow also recovers multiple conformational states from a single input structure. At matched parameter count, its hypergraph representation reaches pairwise-baseline accuracy for long-range collective motions using three- to four-fold less training data. Modeling protein dynamics as multibody processes therefore preserves conformational diversity that pairwise representations tend to smooth away, enabling scalable exploration of function across accessible protein states.